

Session 100%

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Q1. Equilibrium index of an array

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Solved

Stuck somewhere?

Ask for help from a TA and get it resolved.

Get help from TA.

Problem Description

You are given an array **A** of integers of size **N**.
Your task is to find the equilibrium index of the given array
The equilibrium index of an array is an index such that the sum of elements at lower indexes is equal to the sum of elements at higher indexes.
NOTE:
Array indexing starts from 0.
If there is no equilibrium index then return -1.
If there are more than one equilibrium indexes then return the minimum index.

Problem Constraints

1 <= N <= 10⁵
-10⁵ <= A[i] <= 10⁵

Input Format

First argument is an array A .

Output Format

Return the equilibrium index of the given array. If no such index is found then return -1.

Example Input

Input 1:
A=[-7, 1, 5, 2, -4, 3, 0]
Input 2:
A=[1, 2, 3]

Example Output

Output 1:
3
Output 2:
-1

Example Explanation

Explanation 1:
3 is an equilibrium index, because:
A[0] + A[1] + A[2] = A[4] + A[5] + A[6]
Explanation 1:
There is no such index.

See Expected Output

Java7 (Open-Jdk-1.7.0)

```
61  
62  
63  
64  
65  
66  
67  
68  
69  
70  
71  
72
```

lowerIndexSum = prefixSum[input]
higherIndexSum = prefixSum[input]

if(lowerIndexSum == higherIndexSum)
{
 return index;
}

return -1;

Test Output

Code's All Neat!

Compiling your Code...
> Success!

Running Test Cases...
> TestCase - Easy Success
> TestCase - Hard Success

Test

Test With Custom Input

Submit